Physics - Undergraduate Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A meter stick with a speed of 0.8 c moves past an observer. In the observer's reference frame, how long does it take the stick to pass the observer ?

   (a) 1.6 ns
   (b) 2.5 ns
   (c) 4.2 ns
   (d) 6.9 ns

2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?

   (a) 2
   (b) 250
   (c) 5,000
   (d) 10,000

3. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?

   (a) Specific heat
   (b) Thermal conductivity
   (c) Electrical resistivity
   (d) Hall coefficient

4. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is

   (a) 1.2 * 10^9 m/s
   (b) 3.0 * 10^8 m/s
   (c) 1.5 * 10^8 m/s
   (d) 1.0 * 10^8 m/s

5. The negative muon, mu$^{-}, hasproperties most similar to which of the following$?

   6. Electron

   7. Meson

   8. Photon

   9. Boson

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation from a nucleus is typically emitted as visible light.
7. True or False: The Sun's energy primarily comes from thermonuclear reactions between two hydrogen atoms and one helium atom.
8. True or False: A grating spectrometer with a resolving power of 5,000 can resolve wavelengths as close as 1 nm.
9. True or False: A dye laser is the best type of laser for spectroscopy over a range of visible wavelengths.
10. True or False: A single-electron atom with the electron in the l = 2 state has five allowed values for the magnetic quantum number m_l.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the possible transitions for an electron in the n = 4, l = 1 state in hydrogen. Identify and justify which final states are not accessible through allowed transitions.
12. Analyze a one-dimensional collision where a particle of mass 2 m collides with a stationary particle of mass m, sticking together afterward. Calculate the fraction of initial kinetic energy lost and discuss the underlying physical principles.

End of Paper